

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_diversity_residual_replay.
- Best held-out TSPLIB gap: phase7_modular_operator_diversity_residual_replay.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_085349_t
- started_at_local: 2026-05-20 08:53:49
- finished_at_local: 2026-05-20 09:24:56
- duration_hhmm: 00:31
- duration_seconds: 1866.809
- seed_offset: 19000
- replicate_label: t
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.880805	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235667	0.054968	0.121542	0.893186	0.4425	False	0.003387	0.316569
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.824053	0.4475	False	0.009583	0.318133
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.210324	0.041246	0.103538	0.0	0.445	False	-0.003034	2.246598

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.887949	0.445	False	0.006407	1.083761
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.0	0.4475	False	0.009895	0.386275

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.880805, complexity 0.56, and adaptation efficiency -0.028275.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.893186, complexity 0.4425, and adaptation efficiency 0.137352.
- Last-epoch runtime 40.167125 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003387, positive scaffolds 1, Pareto runtime inflation 0.316569.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.22697, "gap_delta": 0.000224, "runtime_inflation": 0.316569, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.824053, complexity 0.4475, and adaptation efficiency 0.009865.
- Last-epoch runtime 93.11585 ms and distance evaluations 6364.75.

- Validation: surviving False, transplant delta 0.009583, positive scaffolds 1, Pareto runtime inflation 0.318133.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.269436, "gap_delta": 0.000224, "runtime_inflation": 0.318133, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.210324, family holdout 0.041246, combined 0.103538.
- Accepted novelty 0.0, complexity 0.445, and adaptation efficiency 0.0.
- Last-epoch runtime 138.293125 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta -0.003034, positive scaffolds 4, Pareto runtime inflation 2.246598.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 26.784097, "gap_delta": -0.01625, "runtime_inflation": 2.246598, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.887949, complexity 0.445, and adaptation efficiency 0.016348.
- Last-epoch runtime 67.91075 ms and distance evaluations 4691.5.
- Validation: surviving False, transplant delta 0.006407, positive scaffolds 2, Pareto runtime inflation 1.083761.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.954353, "gap_delta": -0.003304, "runtime_inflation": 1.083761, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.0, complexity 0.4475, and adaptation efficiency 0.0.
- Last-epoch runtime 39.334 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.009895, positive scaffolds 1, Pareto runtime inflation 0.386275.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.275361, "gap_delta": 0.000224, "runtime_inflation": 0.386275, "same_gap_faster": false}

Judge Appendix

Review of TSP Modular-Operator Discovery Suite Results

Summary of Evaluation Criteria and Priority

- ****Main transfer evidence****: Held-out TSPLIB gap & family holdout gap.
- ****Modular operator interest****: Must survive transplant and ablation validation.
- ****Claims****: Focus on reusable operator structure rather than full solver unless modular conditions strongly support discovery.

- ****Algorithm discovery****: Requires operator survival in scaffold and family tests.

Conditions Overview

Condition Name	Operator Type	Final Family Gap	Final Transfer Gap	Final TSPLIB Gap	Ablation Support	Transplant Mean Gap Delta	Surviving Candidate
phase7_baseline_heuristic_only	n/a	0.054968	0.121317	0.235058	N/A	0.0	No
phase7_full_solver_evolution	n/a	0.064916	0.159398	0.213387	N/A	0.0	No
phase7_modular_operator_evolution	candidate_pruner	0.054968	0.121542	0.235667	No	0.003387	No
phase7_modular_operator_random_replay	candidate_pruner	0.054968	0.121542	0.235667	No	0.009583	No
phase7_modular_operator_diversity_residual_replay	scaffold_selector	0.041246	0.103538	0.210324	Partial (ablation shows disabling increases gap)	-0.003034	No
phase7_modular_operator_compression_pressure	restart_controller	0.050827	0.118702	0.235058	No	0.006407	No
phase7_modular_operator_pareto_selection	candidate_pruner	0.054968	0.121542	0.235667	No	0.009895	No

Detailed Interpretation

1. Held-Out Transfer Gap and Family Holdout Gap

- The lowest held-out TSPLIB gap (0.210324) and transfer gap (0.103538) are reported by ****phase7_modular_operator_diversity_residual_replay (scaffold_selector)****.
- This condition also shows ****best family-level transfer improvements**** with:
 - Mean family gap delta $\approx$ -0.0153 (improvement)
 - Positive scaffolds count = 4 (best positive scaffold transplant count)
 - Helpful families: heldout_tsplib, clustered_tsp, grid_like_tsp
 - However, transplant results show:
 - Does ***not*** transplant cleanly into cheapest_insertion_2opt.
 - Runtime inflation is very high (2.2466 $\times$).
 - Ablation:

- Disabling operator ****increases gap by +0.016**** → moderate ablation signal supporting an operator effect.
- However, a variant ("flattened_selector") performs even worse (+0.048 gap), while "shuffled_selector" slightly improves (-0.009).
- Despite good transfer & family gap results, the operator ****did not survive as candidate****.

2. Modular Candidate Pruners (3 variant conditions: evolution, random_replay, pareto_selection)

- Final TSPLIB gaps ~0.2356, transfer gaps ~0.1215 — close to baseline.
- Ablation shows ****no significant penalty when disabling**** operator (disabling gap delta slightly negative ~ -0.0002), i.e., no ablation support.
- No family shows clear improvement; positive family count = 0.
- Runtime inflation ~0.3× (significant overhead).
- Transplant results:
 - Mean transplant gap delta slightly ****worse**** than baseline (0.0034 to 0.0099).
 - Only one or zero positive scaffolds.
 - Not surviving candidate status.
- Conclusion: No evidence of reusable operator structure validated by ablation or transplant.

3. Restart Controller (phase7_modular_operator_compression_pressure)

- Moderate final transfer gap (0.1187) slightly better than baseline (0.1213).
- Ablation: disabling operator slightly worsens gap (+0.0033), which is weak ablation signal.
- Family-level signals weakly positive but only for one family (two_cluster_bottleneck_tsp).
- Runtime inflation ~1.08× and does not transplant cleanly into some scaffolds.
- Transplant mean gap delta positive (worsening) 0.0064 with only 2 positive scaffolds.
- Not surviving candidate status.

Pareto Tradeoffs

- Scaffold selector (best transfer gap) has the highest runtime inflation (2.25×), but reduces gap more than others (~-0.01625 gap delta vs baseline).
- Candidate pruners offer minor or no gap improvements with ~0.3× runtime inflation.
- Restart controller has small gap improvement (-0.0033) with ~1.08× runtime inflation.
- No operator achieves better gap without runtime inflation or better runtime without losing gap.

Transplant Validation

- No operator cleanly transplants across all tested scaffolds.

- Scaffold selector has positive scaffold counts but introduces large runtime overhead and inconsistent transplant success (negative transplant gap delta for some scaffolds).
- Candidate pruners and restart controllers show modest or negative transplant results with no ablation validation.

Ablation Summary

- Scaffold selector is the only operator with a moderate ablation signal indicating it contributes positively to solution quality.
- Candidate pruners and restart controllers fail ablation tests (disabling does not degrade performance), so their effect is not validated.

Conclusion and Recommendations

Operator	Considered Interesting?	Reasoning
Scaffold Selector (phase7_modular_operator_diversity_residual_replay)	Marginally interesting but not surviving	Shows best transfer family gap improvements and some ablation support; however, large runtime inflation and failure to survive transplant and family validations prohibit claim of algorithm discovery.
Candidate Pruners (modular operator conditions)	Not interesting	No ablation validation, no family or transplant improvement, runtime inflation ~30%.
Restart Controller	Not interesting	Weak ablation, limited family help, runtime inflation >1×, no transplant survival.

No modular operator survived the full validation pipeline to support claims of reusable operator structure or algorithm discovery.

Final Summary

- **No modular operator qualifies as a surviving candidate.**
- The **scaffold selector operator** (phase7_modular_operator_diversity_residual_replay) shows the strongest evidence of reusable structure via transfer gap and family improvements with some ablation signal but suffers from large runtime inflation and transplant issues. It remains a **potentially reusable operator** but not confirmed.
- Candidate pruners and restart controllers fail ablation and transplant validation; their minor gap differences are indistinguishable from noise.
- Pareto tradeoffs disfavor these operators because improvements come at high runtime cost without robust validation.
- Full-solver evolution and baseline produce higher transfer gaps, indicating modular approaches capture some structural signal but fall short of validated improvement.
- **No claims of algorithm discovery or reliably reusable operator structure can be made.**

Recommendations

- Focus future modular operator discovery on improving ablation and transplant robustness.
- Reduce runtime inflation to increase Pareto competitiveness.
- Scaffold selector variants merit further enhancement due to best transfer/family signals but require runtime and transplantation fixes.
- Candidate pruners and restart controllers need redesign or stronger validation signals before acceptance.

This interpretation is conservative and prioritizes held-out gap, ablation, family transfer, transplant, and Pareto metrics in accordance with specified rules.